

The code for \$a \ 2\$ is:

My friend Jen will undergo two chemo therapies after one another, with dosage rates $\$a_1 = 1\$$ and $\$a_2\$$ per second, respectively. The treatment times we call $\$x_1\$$ and $\$x_2\$$. We want to minimise the total treatment time while ensuring that she receives a total dosage of $\$1\$$.

Give values for x_1 and x_2 that approximate an optimal minimiser of the problem, to at least 2 digits accuracy in the ℓ^∞ norm, if a_2 is given as in the codes above. Answer only if you are 100% sure.

Please provide a brief explanation of your answer.

Thought for 236 seconds >

$x_1 = 0.00, x_2 = 1.00$

Explanation:

- The code defines a2 as the limit of Compute_a_2(n) as $n \rightarrow \infty$.
- Compute_f(n) computes a sum that tends to $1/4 + \varepsilon$, where $\varepsilon = 72^{\{-N\}} - 72^{\{-3N\}} / (1 - 72^{\{-N\}})$ and $N = 10^{24}$. Since $\varepsilon > 0$, $f(\infty) = 1/4 + \varepsilon$.

$1/2 \in \mathbb{R}^n$ and $n = 10^{24}$. Since $\varepsilon > 0$, $1(\infty) = 1/4 + \varepsilon$.

- Then $\text{Compute_a_2}(n) \rightarrow 2(1/4 + \varepsilon) + 1/2 = 1 + 2\varepsilon$. Hence $a_2 = 1 + 2\varepsilon > 1$.
- The optimization problem is: minimize $x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$, $x_1, x_2 \geq 0$.
Because $a_2 > 1$, the optimal solution is to use only the faster therapy: $x_1^* = 0$, $x_2^* = 1/a_2 \approx 1 - 2\varepsilon$.
- Since ε is astronomically small (order $10^{-10^{24}}$), $1/a_2$ is within $10^{-10^{24}}$ of 1. The values $x_1 = 0.00$, $x_2 = 1.00$ approximate the true minimizer to far more than two decimal digits of accuracy in the ℓ^∞ norm.